Computer Science & IT



Data Structure

Tree

Lecture No. 07



By- Abhishek Sir

Recap of Previous Lecture



Topic	BST, Deletion (predecessor, Successor)
Topic	No. of BST
Topic	AVL. balancing factor
Topic	Critical Node
Topic	Rotation

Slide

Topics to be Covered



- Topic — RR Rotation
- Topic — LR Rotation
- Topic — RL. Rotation
- Topic — Minimum No. of Node in BST with height h
- Topic — Heap

Slide



Topic : Tree

AVL Tree Minimum number of nodes
HEAP
Insert in MAX HEAP

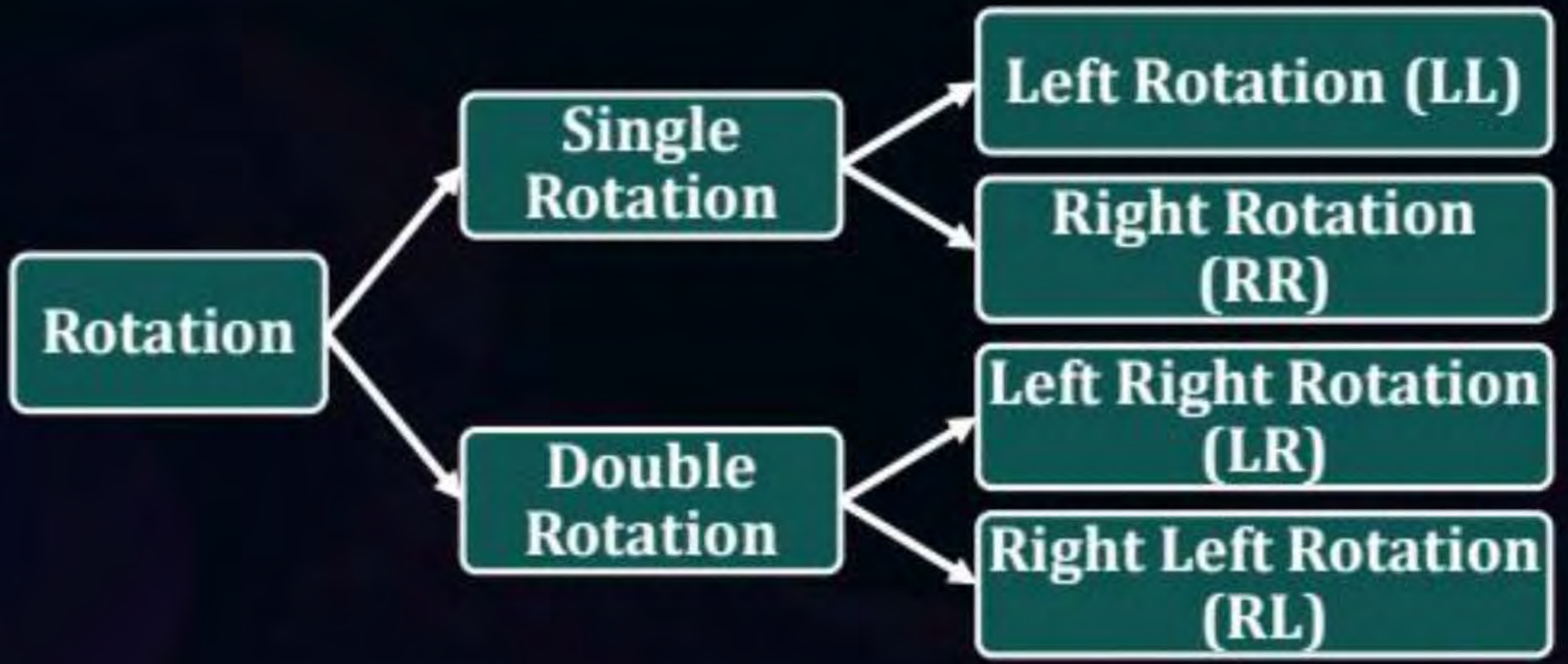
Slide



Topic : AVL Tree

Saturday

To balance itself, an AVL tree may perform the following four kinds of rotations



Right Rotation

left Rotation

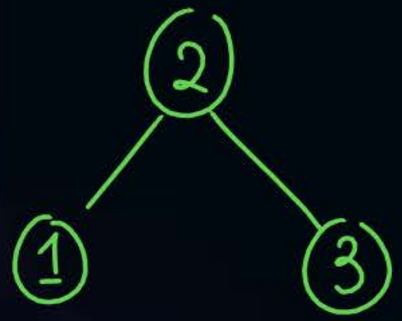


Topic : AVL Tree



2 edges indentify critical Node to Newly inserted Node

LL Rotation Right Rotation



Slide



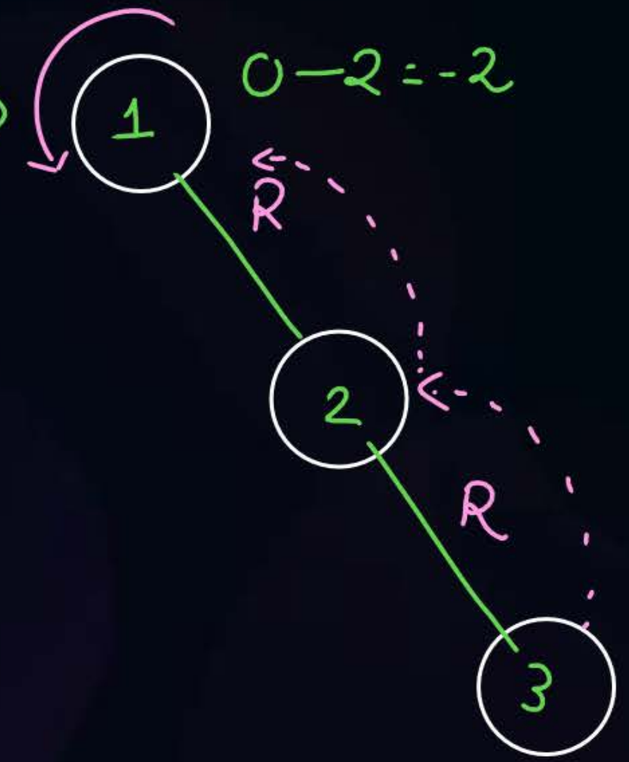
Topic : AVL Tree

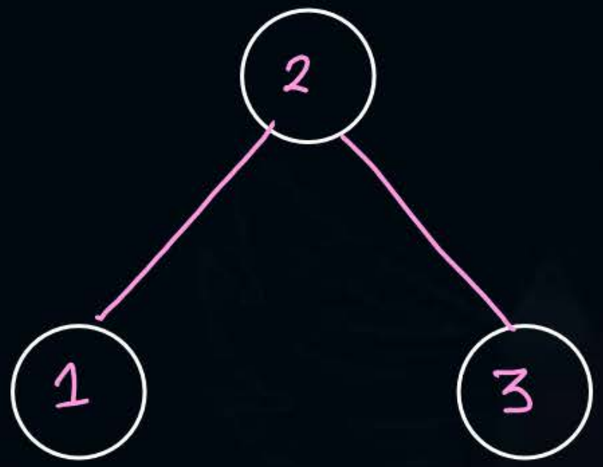
RR Rotation

RR Rotation → 1 Left Rotation

Insert 1, 2, 3

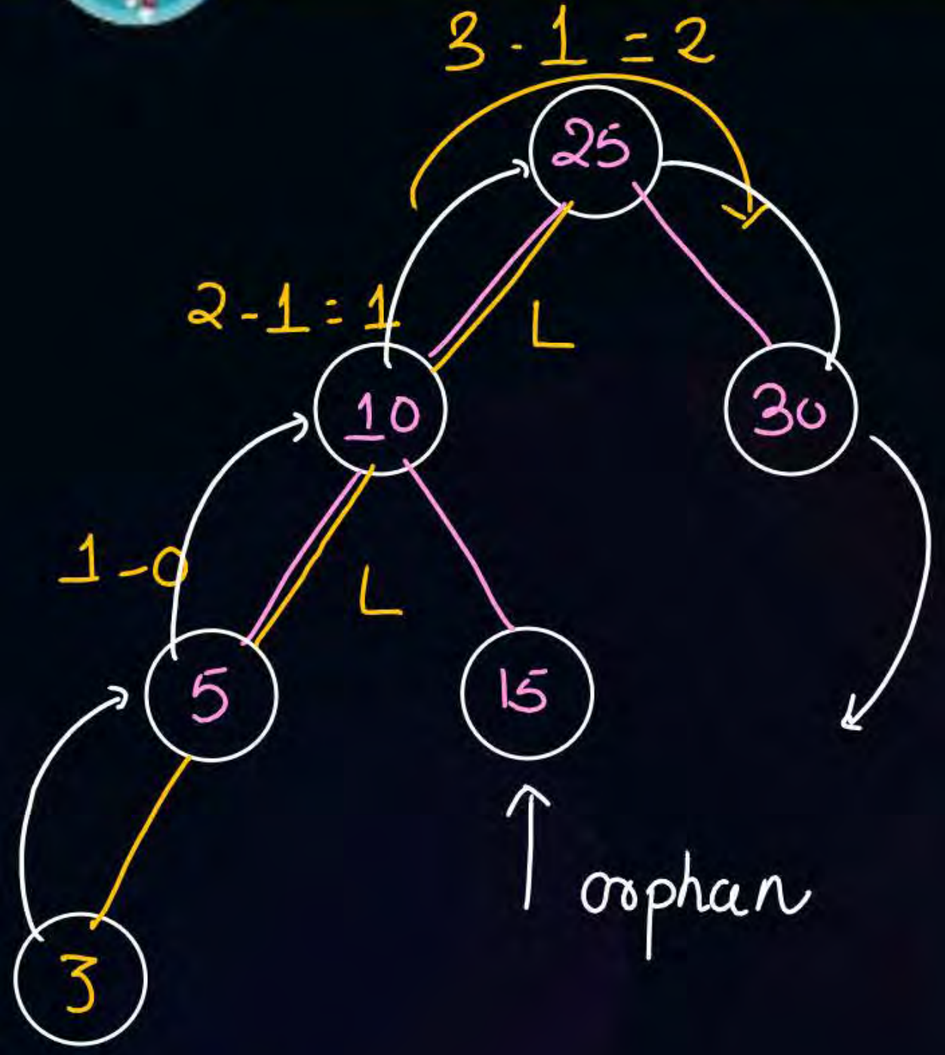
Critical Node ⟹





Slide

Topic : AVL Tree



Insert 3

Critical Node : - 25

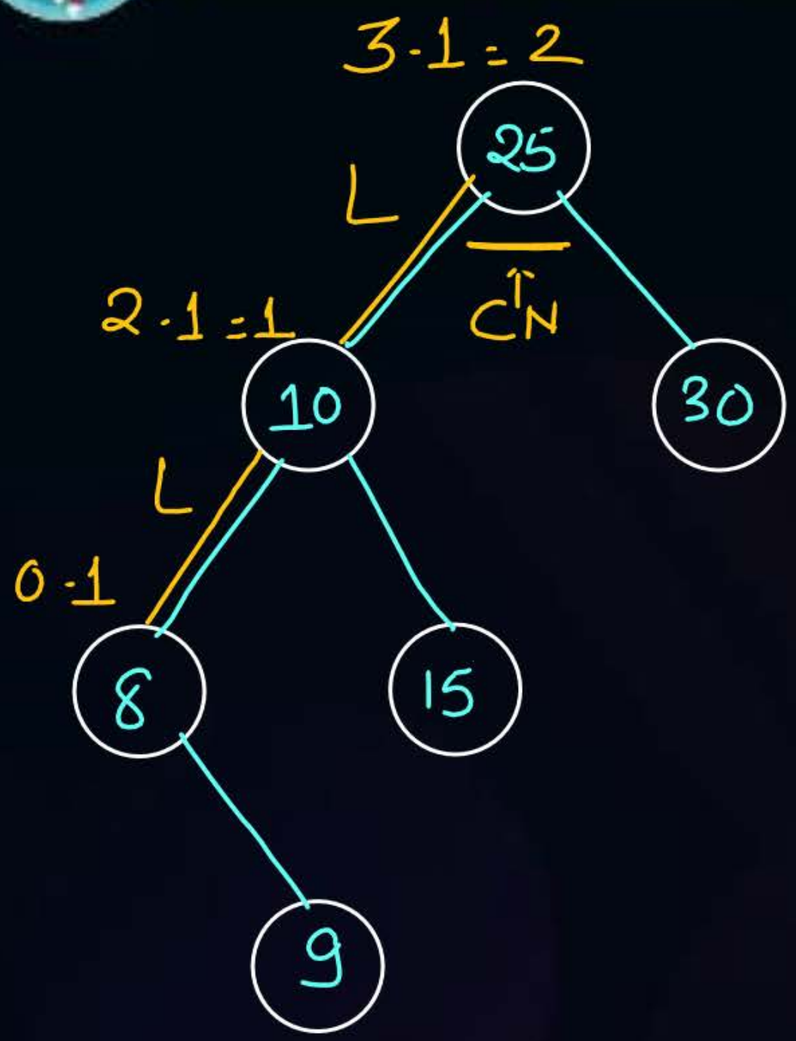
Two edges : LL Rotation

15 is on left subtree of critical Node after rotation it will become left child of critical Node



Slide

Topic : AVL Tree



Insert 9

CN

LL.



Slide



Topic : AVL Tree

Insert 3, 1, 2

Critical Node 3



Two edges :- LR Rotation Double Rotation

L Left child of critical Node Rotated left

R Critical Node Rotated Right

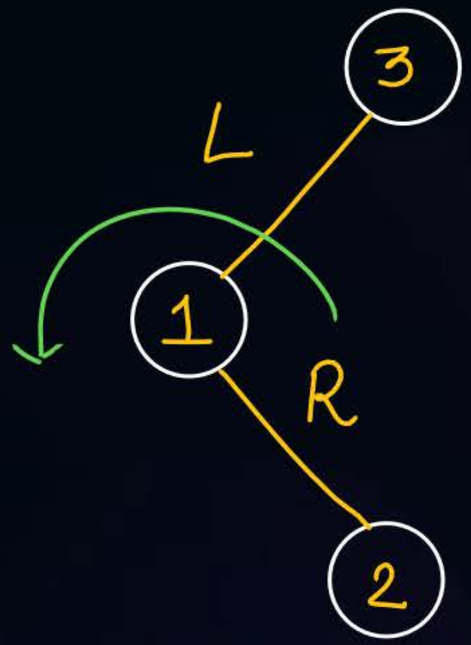
Slide

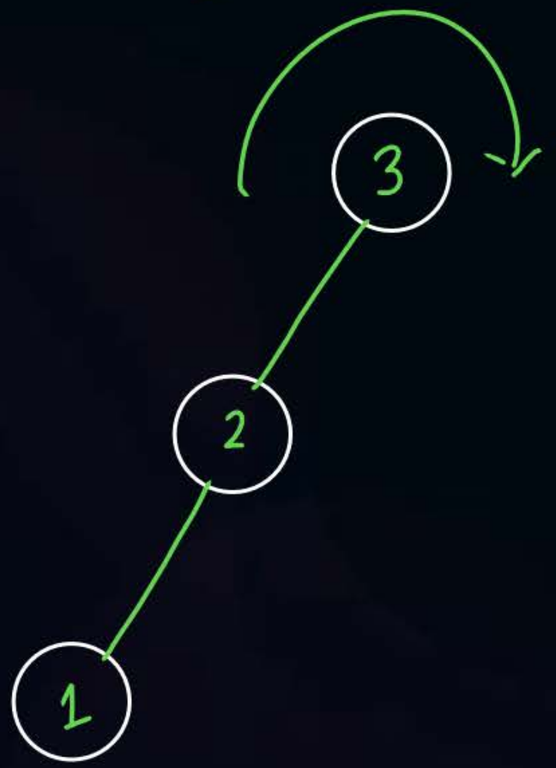


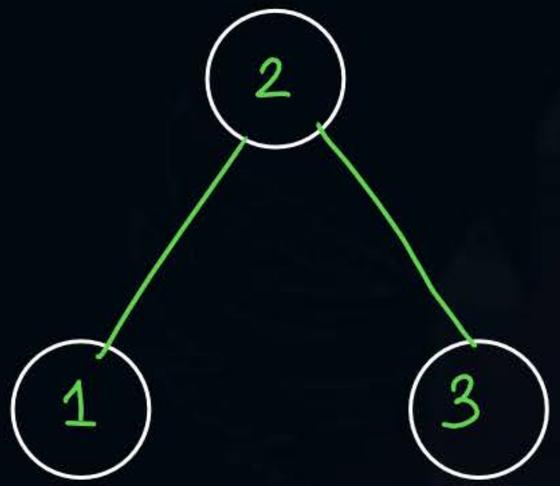
Topic : AVL Tree

Insert 3, 1, 2

Critical Node 3





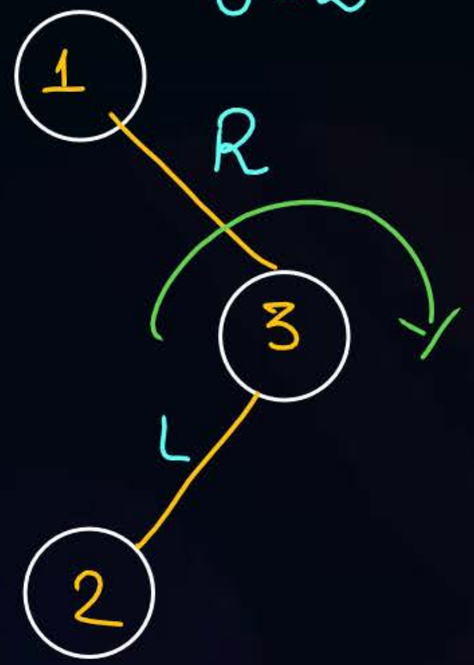


Slide

Topic : AVL Tree

Insert 1, 3, 2

0 - 2



Critical Node - 1

Two edges :- R - L

R : Right child of critical Node

Rotated right

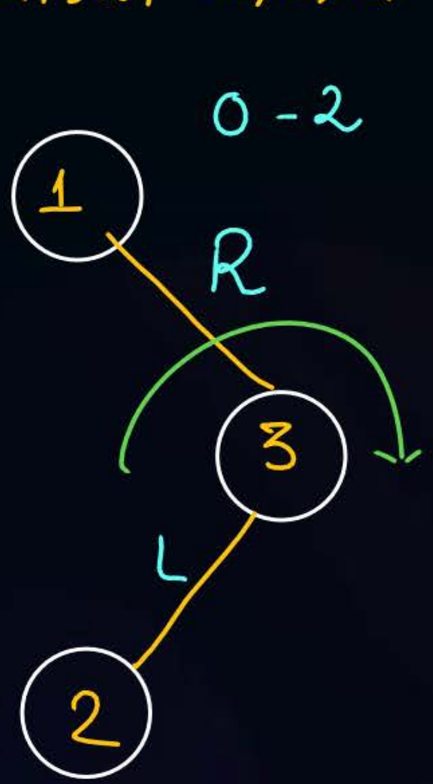
L. critical Node Rotated left

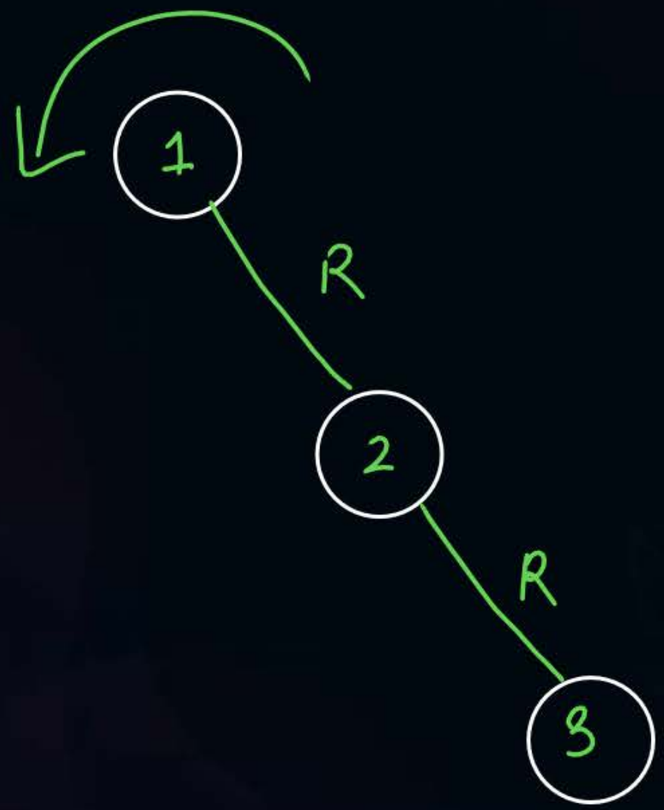
Slide

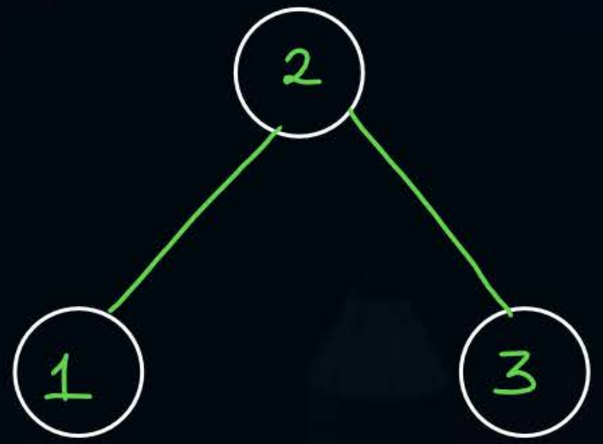


Topic : AVL Tree

Insert 1, 3, 2







Slide



Topic : AVL Tree

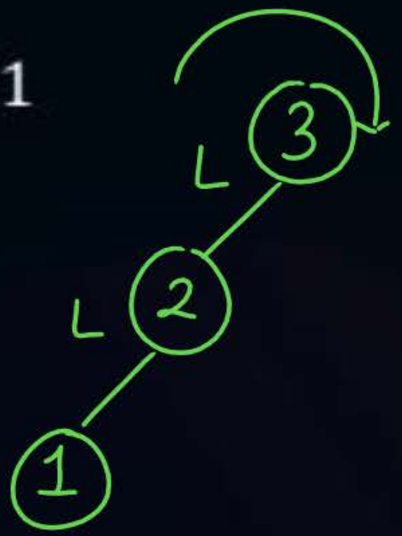
Insert the following sequence of element in AVL tree

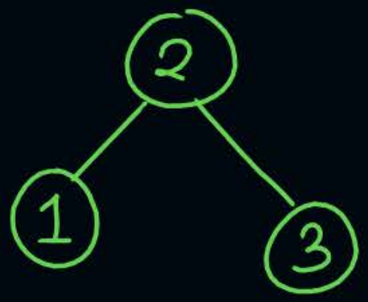
3, 2, 1, 4, 5, 6, 7, 15, 16, 14, 13



Topic : AVL Tree

Insert 3, 2, 1







Topic : AVL Tree

Insert the following sequence of element in AVL tree

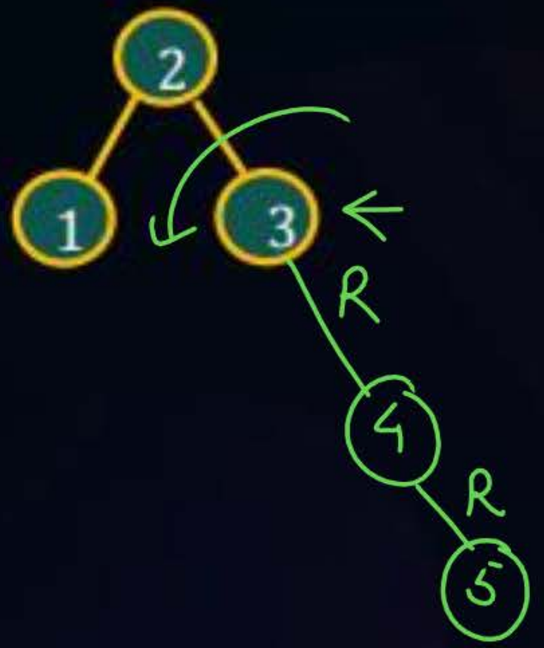
3, 2, 1, 4, 5, 6, 7, 15, 16, 14, 13

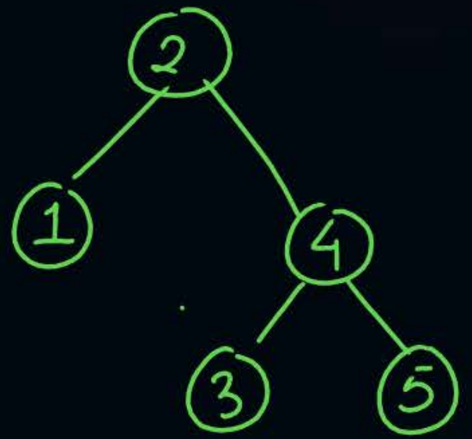




Topic : AVL Tree

Insert 4, 5

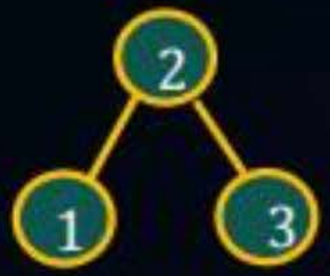


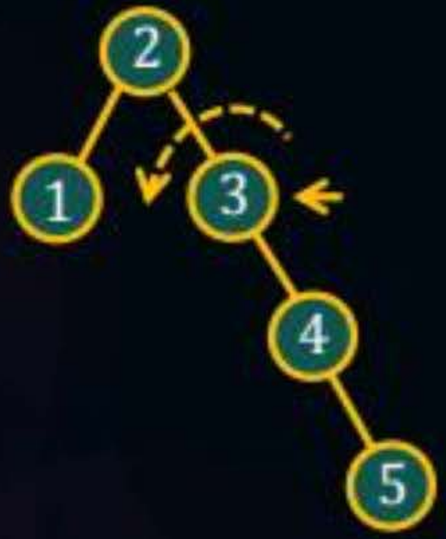




Topic : AVL Tree

Insert 4, 5

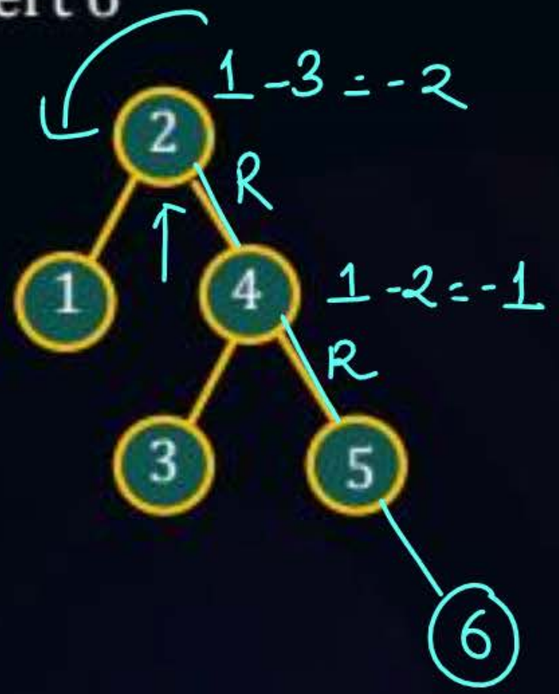






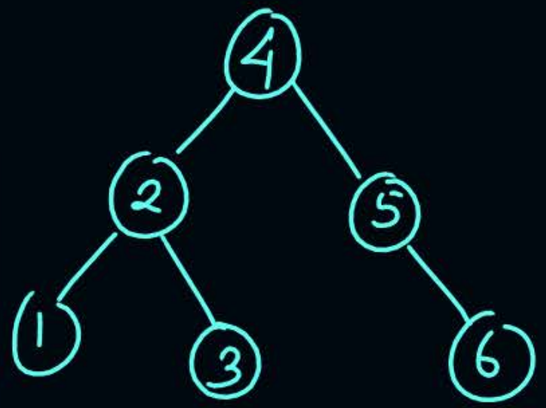
Topic : AVL Tree

Insert 6



Critical Node :

Type of Rotation :

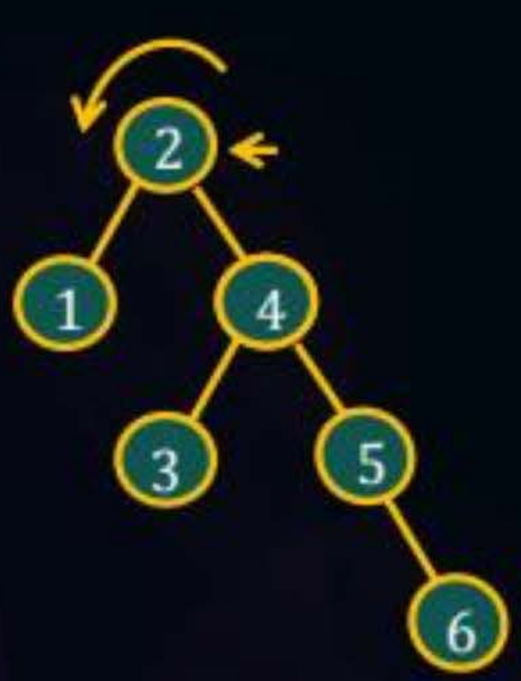


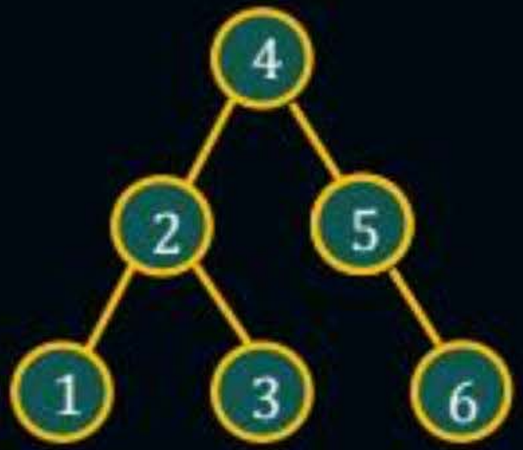
3 is in Right subtree of critical Node after rotation right child of critical Node



Topic : AVL Tree

Insert 6

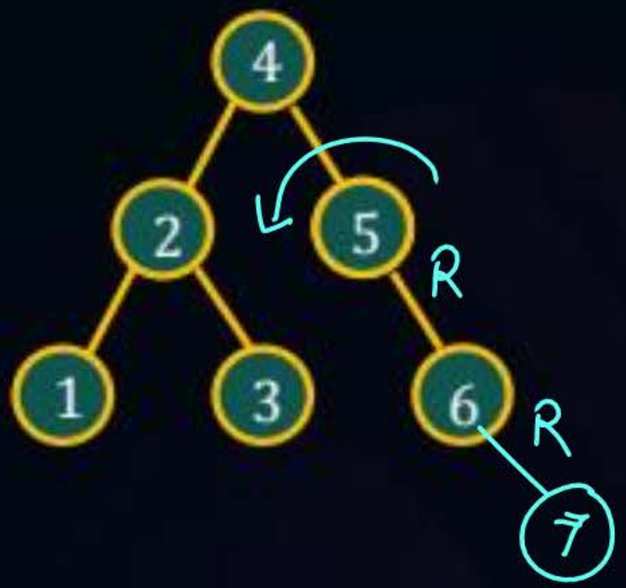


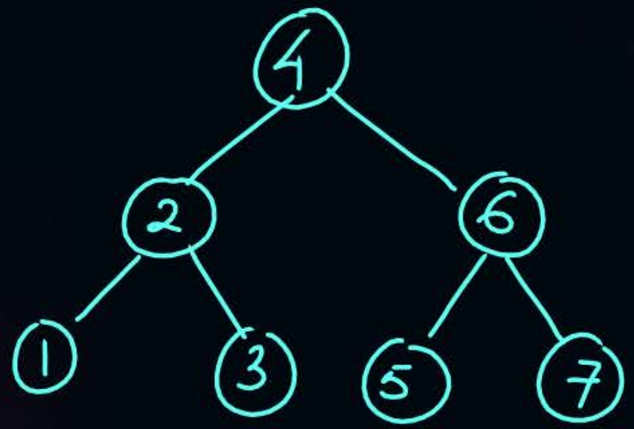




Topic : AVL Tree

Insert 7

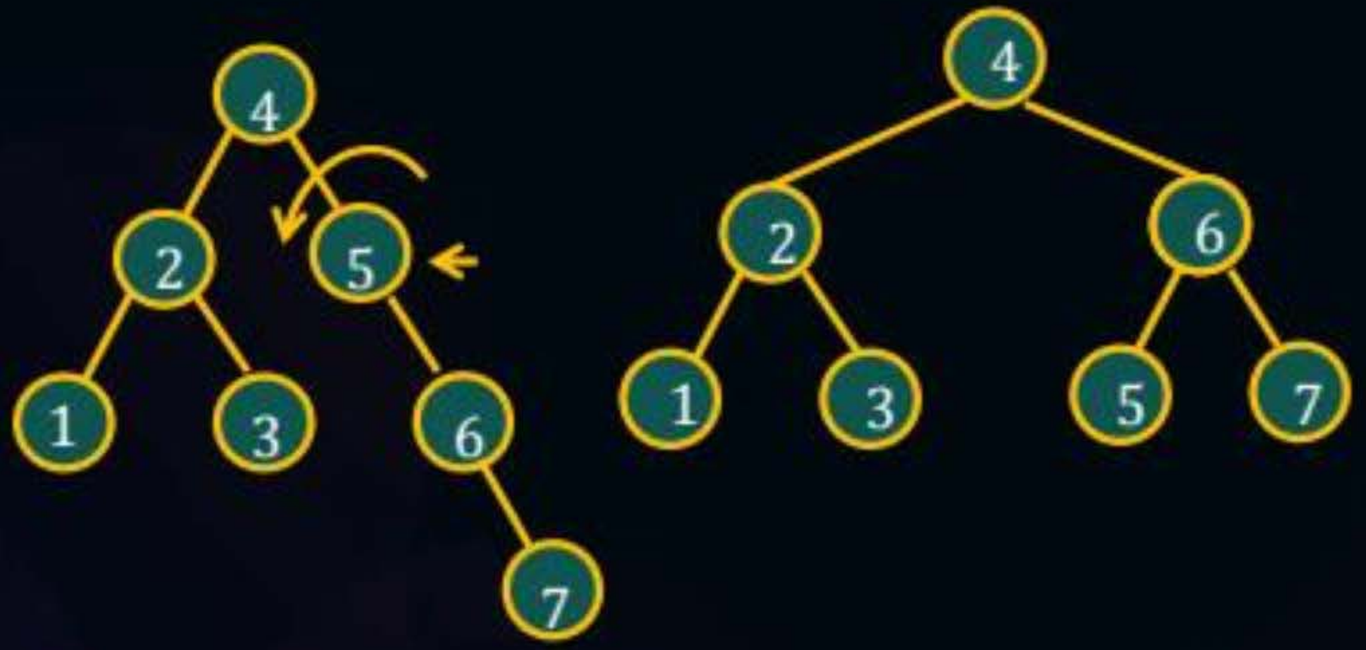






Topic : AVL Tree

Insert 7





Topic : AVL Tree

Insert 15, 16





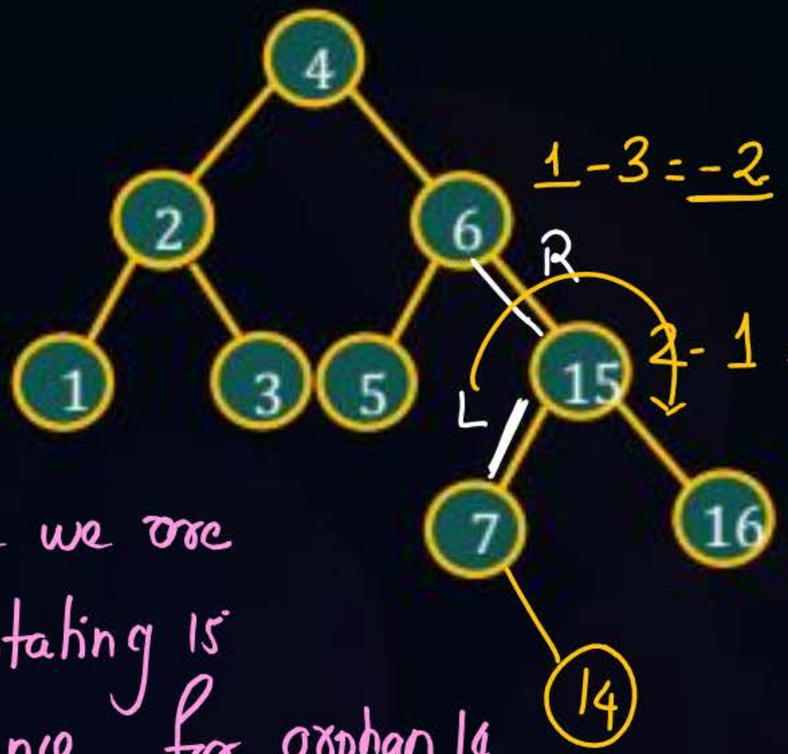
Topic : AVL Tree

Insert 15, 16



Topic : AVL Tree

Insert 14



Concentrate

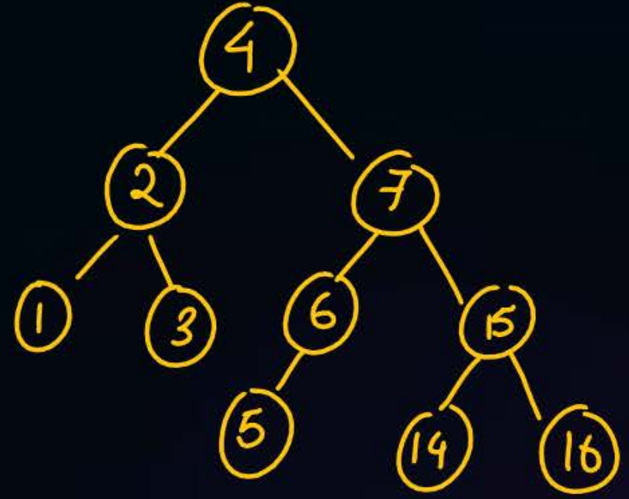


14 Insert

Critical Node

Type of Rotation RL

⇒



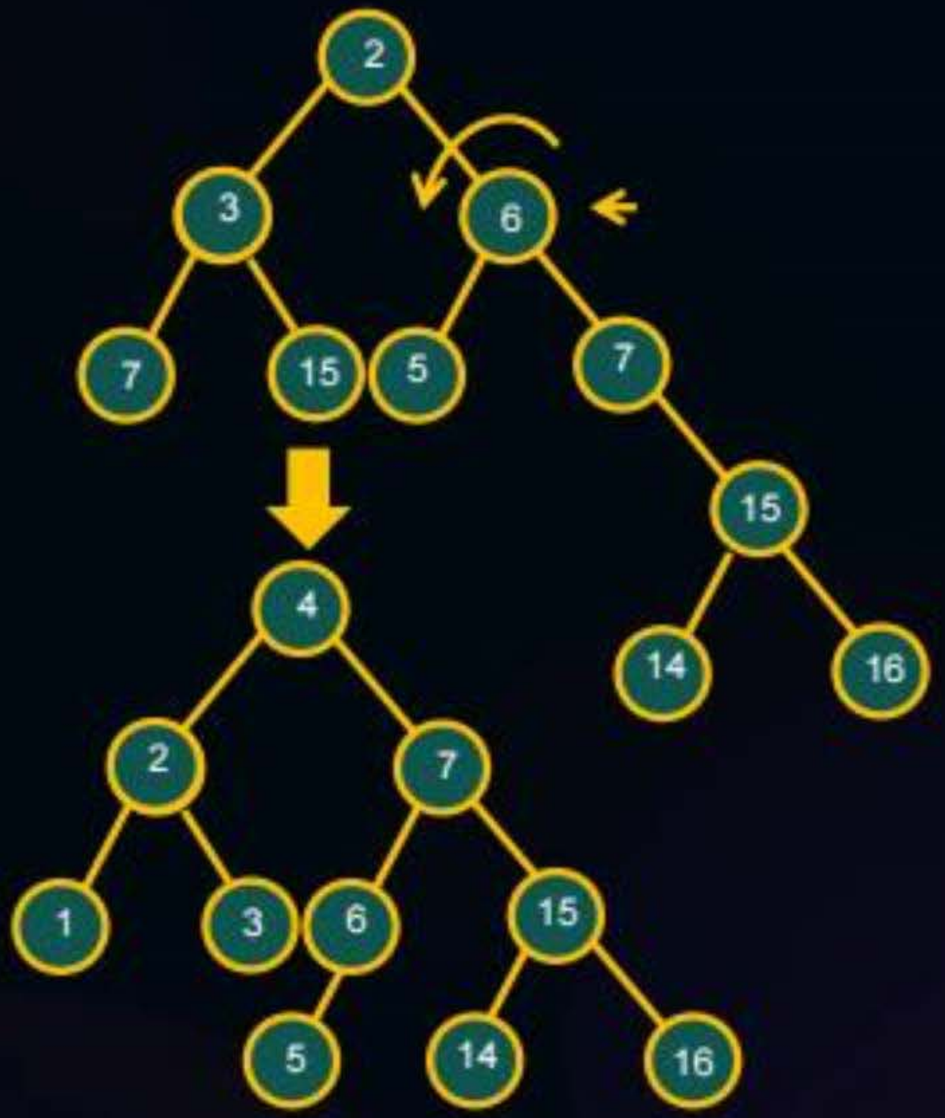
Since we are Rotating is Hence for orphan 14 we look for 15 only



Topic : AVL Tree

Insert 14

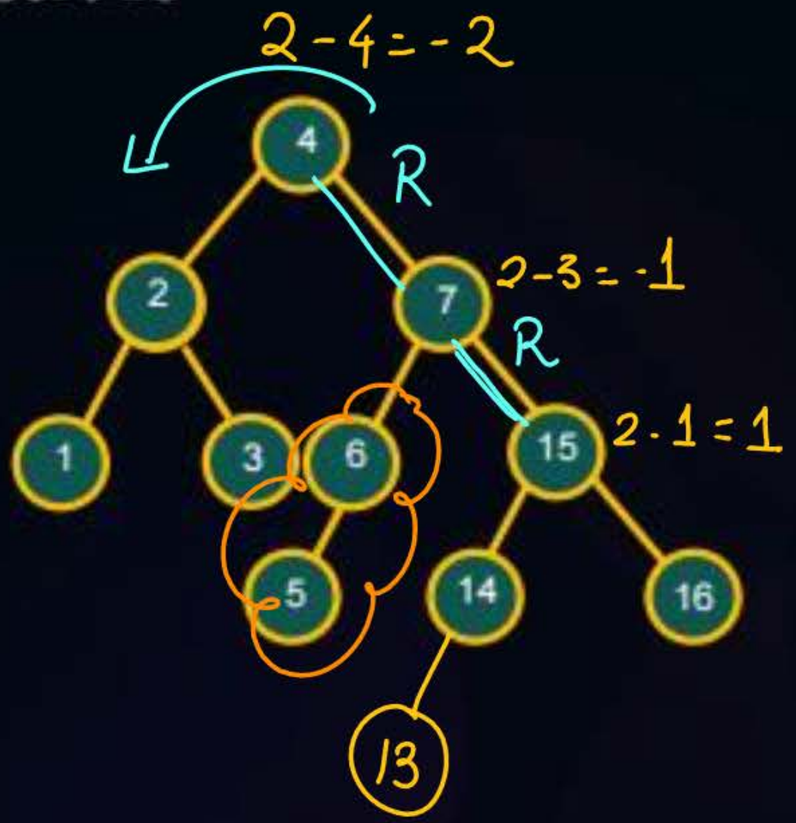


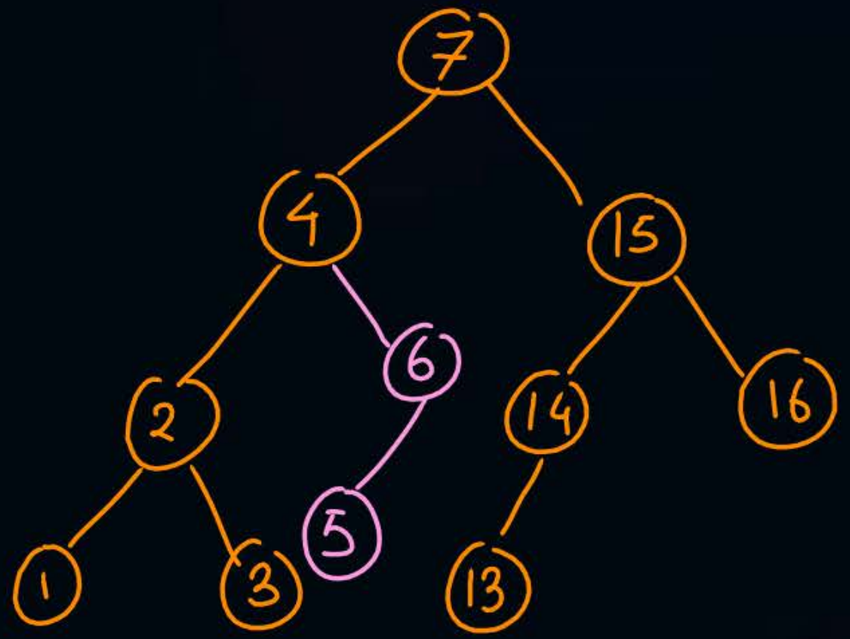




Topic : AVL Tree

Insert 13

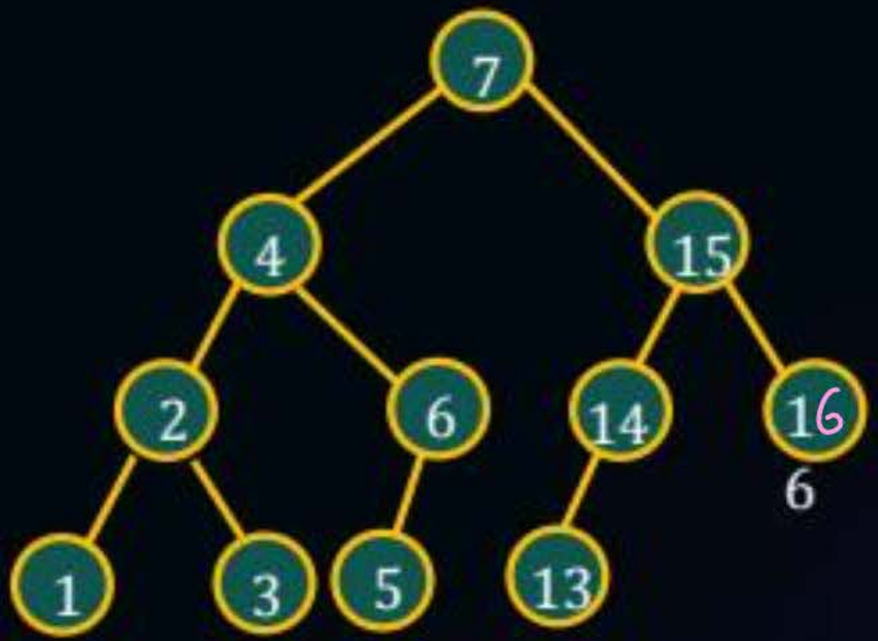




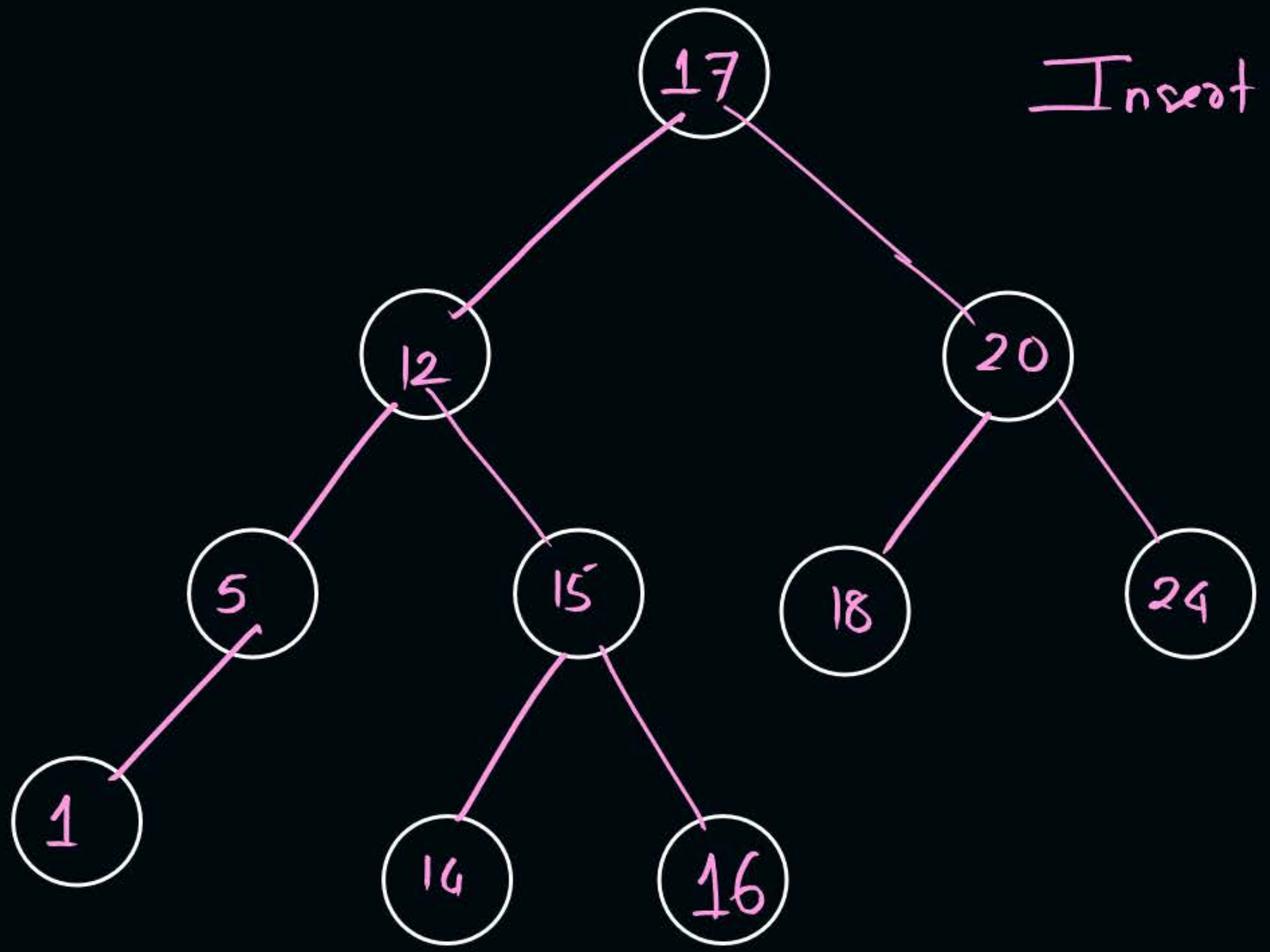
Insert 13

AVL insertion





6



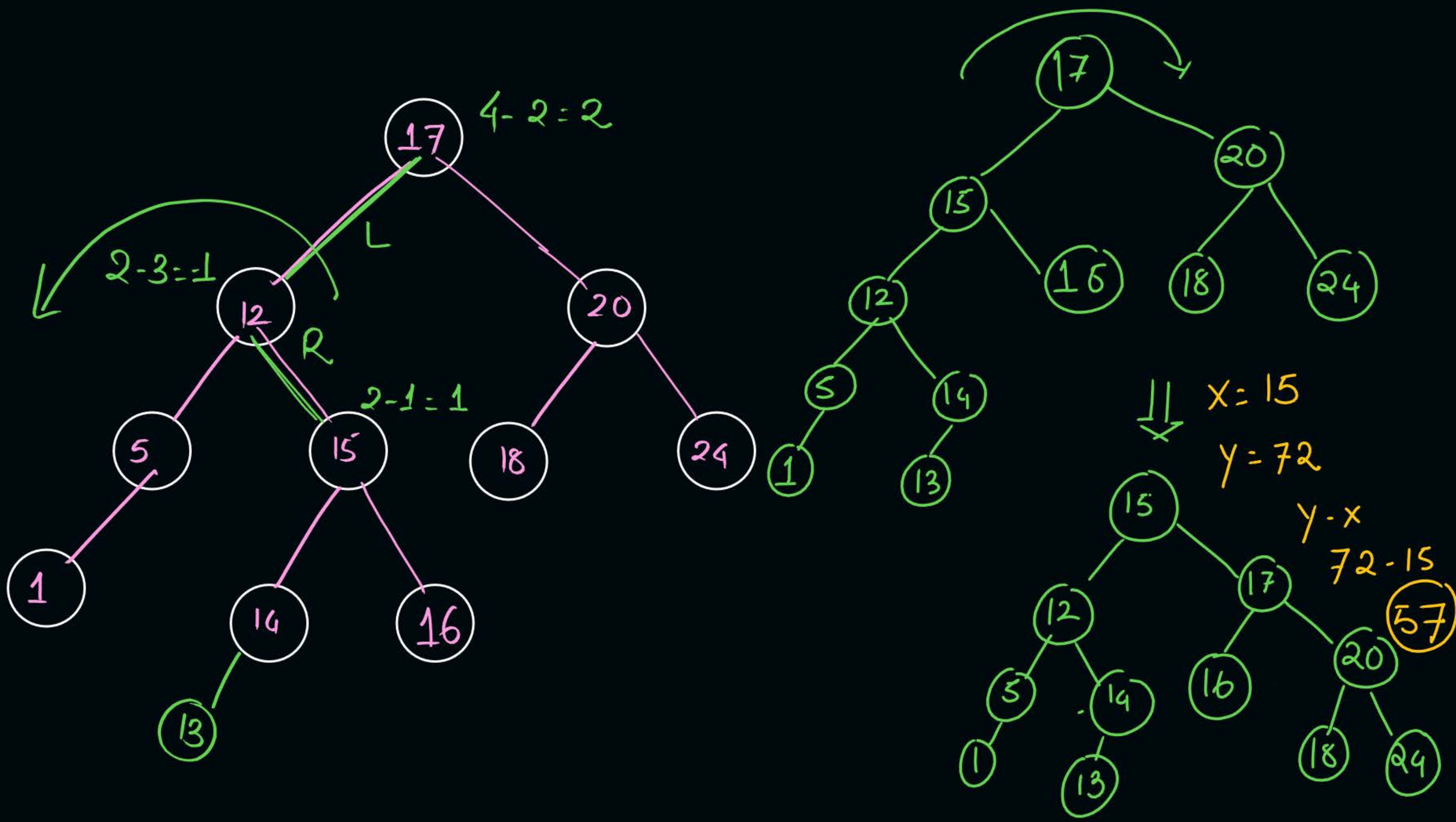
Insert 13 in AVL tree

if Root node x and

and Sum of leaf Node is y

then y − x is [57]

Calculation strong


17
4-2=2
L
2-3=-1
12
R
20
2-1=1
5
15
18
24
1
14
16
13
17
20
15
12
16
18
24
5
14
1
13
x=15
y=72
y-x
72-15
57
15
17
12
20
5
14
16
1
13
18
24



Topic : AVL Tree

Operation	Cost of the operation
Search	$O(\log_2 n)$
Minimum	$O(\log_2 n)$
Maximum	$O(\log_2 n)$
Predecessor	$O(\log_2 n)$
Successor	$O(\log_2 n)$
Insert	$O(\log_2 n)$
Delete	$O(\log_2 n)$
BST Construction	$O(n\log n)$

$\log_2 n$

Height



Topic : AVL Tree

Minimum No. of Nodes in AVL tree of height h $n(h)$

1. $h = 0$



$n(0) = 1$

2. $h = 1$



$n(1) = 2$

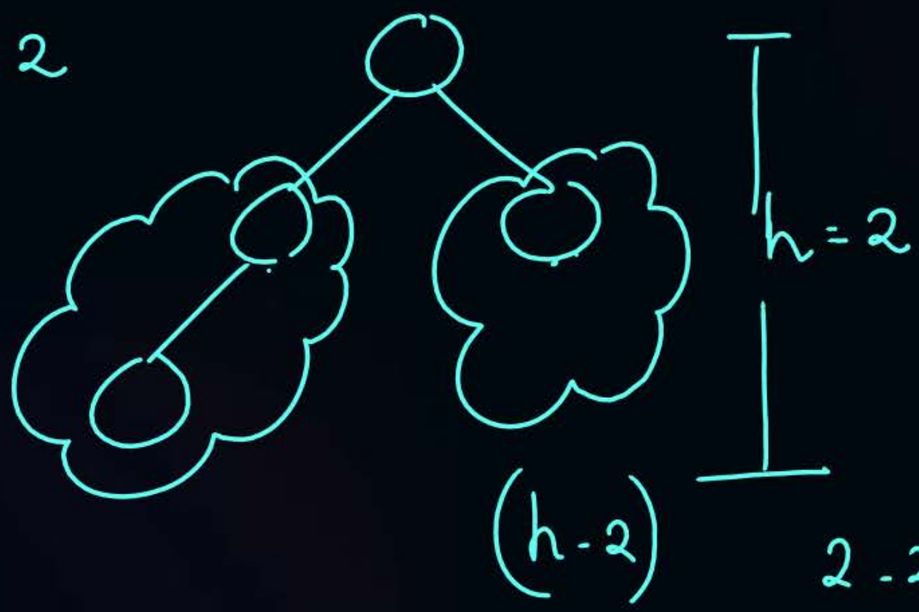
Minimum ✓

Minimum ✗

Slide

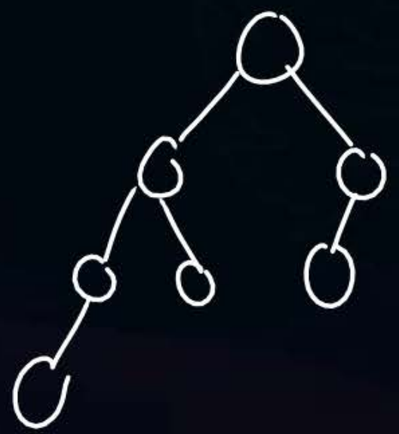
Topic : AVL Tree

3 $h=2$



$n(2) = 4$

$2-2=0$



4. $h=3$



$h(3) = 7$ ✓

$n(4) = n(3) + n(2) + 1$

$7 + 4 + 1 = 12$

Slide



Topic : AVL Tree

$$n(h) = n(h-1) + n(h-2) + 1$$ ← Root

$n(5) = n(4) + n(3) + 1$

$12 + 7 + 1 = 20$

Slide



Topic : AVL Tree

#Q If No of AVL tree with 1,2,3,4 is x and Minimum No of Node in AVL tree of Heigh 8 is y then x+y is ——

[88]

$n(3) = 7$

$n(4) = 12$

$n(5) = 20$

$n(6) = 33$

$n(7) = 54$

$n(8) = 88$

88+4 = 92

Slide



Topic : AVL Tree

with n distinct element
BST with height n-1 is 2^{n-1}

4 Node unlabeled binary tree



Slide



2 mins Summary

Topic AVL tree Rotation

Topic RR

Topic LR

Topic RL

Topic Minimum No of Node in Height h

$$n(h) = n(h-1) + n(h-2) + 1$$

$$n(0) = 1$$

$$n(1) = 2$$

Slide

THANK - YOU

Slide